

Best Subset for Predicting Private Undergraduate Institution In-State Tuition

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Read in Data

Read csv from this link into a dataframe `df`.

Code book available here.

Each row is unique by column UNITID.

1. Data Cleaning

For this analysis, we will focus on only Private school.

```
##           CONTROL  min  median   max      sd  n missing
## 1 Private for-profit 5160 16449.0 51400 7555.020 80      14
## 2 Private not-for-profit 0 38400.0 85000 16047.532 778      6
## 3 Public           910  7019.5 21989  3804.278 758      4
```

```
## Number of Private schools = 858
```

```
## Number of columns = 122
```

1st Cleaning

The next code does the following.

- filter the schools down to only Private for-profit and Private not-for-profit
- create dummy variables to distinguish between for-profit and not-for-profit
- turn any categorical variables into dummy variables (if there are n levels, there will be $n-1$ extra variables)
- remove the original categorical variables because we will use the created dummies for analysis from now on.

```
private1 <- df %>%
  filter(CONTROL %in% c("Private for-profit", "Private not-for-profit")) %>%
  # categorical variables (if there are n levels, there will be n-1 columns)
  mutate(
```

```

FORPROFIT = ifelse(CONTROL == "Private for-profit", 1, 0),
ADMCON7_REQUIRED = ifelse(ADMCON7 == "Required to be considered for admission", 1, 0),
APPFEEUW_YES = ifelse(APPFEEUW == "Yes", 1, 0), # application fee waiver
TUITPL1_YES = ifelse(TUITPL1 == "Yes", 1, 0), # tuition guaranteed plan
TUITPL3_YES = ifelse(TUITPL3 == "Yes", 1, 0), # tuition payment plan
ALLONCAM_YES = ifelse(ALLONCAM == "Yes", 1, 0), # Stay on-campus required
ASSOC1_YES = ifelse(ASSOC1 == "Yes", 1, 0), # member of NCAA
BOARD_YES = ifelse(BOARD != "No", 1, 0), # provide meal plan
ROOM_YES = ifelse(ROOM == "Yes", 1, 0), # provide housing
SEMESTER = ifelse(CALSYS == "Semester", 1, 0), # calendar system
HBCU_YES = ifelse(HBCU == "Yes", 1, 0), # Historically Black
HOSPITAL_YES = ifelse(HOSPITAL == "Yes", 1, 0), # has hospital
OPENPUBL_YES = ifelse(OPENPUBL == "Institution is open to the public", 1, 0),
RELAFFIL_NO = ifelse(RELAFFIL == "Not applicable", 1, 0), # not religious
SLO6_YES = ifelse(SLO6 == "Yes", 1, 0), # has study abroad
SLOA_YES = ifelse(SLOA == "Yes", 1, 0), # has undergraduate research
STUSRV2_YES = ifelse(STUSRV2 == "Yes", 1, 0), # has career service
STUSRV3_YES = ifelse(STUSRV3 == "Yes", 1, 0), # work-study to pay tuition

# LOCALE: degree of urbanization
## not include Rural
LO_SUBURB = ifelse(LOCALE %in% c("Suburb: Small", "Suburb: Large", "Suburb: Midsize"), 1, 0),
LO_TOWN = ifelse(LOCALE %in% c("Town: Distant", "Town: Remote", "Town: Fringe"), 1, 0),
LO_CITY = ifelse(LOCALE %in% c("City: Midsize", "City: Small", "City: Large"), 1, 0),

# HDEGOFR1: Highest degree offered
## not include Bachelor
HDEOFR_DOC = ifelse(HDEGOFR1 %in% c("Doctor's degree - research/scholarship", "Doctor's degree - pro
HDEOFR_MAS = ifelse(HDEGOFR1 %in% c("Master's degree"), 1, 0),

# OBEREG: Bureau of Economic Analysis (BEA) regions
## not include U.S. Service schools
NORTHEAST = ifelse(OBEREG %in% c(
  "New England (CT, ME, MA, NH, RI, VT)",
  "Mid East (DE, DC, MD, NJ, NY, PA)"), 1, 0),
MIDWEST = ifelse(OBEREG %in% c(
  "Great Lakes (IL, IN, MI, OH, WI)",
  "Plains (IA, KS, MN, MO, NE, ND, SD)"), 1, 0),
SOUTH = ifelse(OBEREG %in% c(
  "Southeast (AL, AR, FL, GA, KY, LA, MS, NC, SC, TN, VA, WV)",
  "Southwest (AZ, NM, OK, TX)"), 1, 0),
WEST = ifelse(OBEREG %in% c(
  "Far West (AK, CA, HI, NV, OR, WA)",
  "Rocky Mountains (CO, ID, MT, UT, WY)"), 1, 0),

# CARNEGIE CLASSIFICATION
SZ_MEDIUM = ifelse(CARNEGIESIZE %in% c("Medium"), 1, 0),
SZ_LARGE = ifelse(CARNEGIESIZE %in% c("Large", "Very Large"), 1, 0),
RSCH_INSTS = ifelse(CARNEGIERSCH == "Research Colleges and Universities", 1, 0),
RSCH_HIGH = ifelse(CARNEGIERSCH %in% c("Research 1: Very High Spending and Doctorate Production", "I
## leave out low
MED_HI_EARNS = ifelse(CARNEGIESAEC %in% c("Lower Access, Medium Earnings", "Lower Access, Higher Ea
## APM: Program Mixes

```

```

MIXED = ifelse(CARNEGIEAPM == "Mixed", 1, 0),
PROFESSIONS = ifelse(CARNEGIEAPM == "Professions-focused", 1, 0),
HEALTH = ifelse(CARNEGIEAPM %in% c(
  "Special Focus: Nursing",
  "Special Focus: Other Health Professions",
  "Special Focus: Medical Schools and Centers"
), 1, 0),
ARTS = ifelse(CARNEGIEAPM %in% c(
  "Special Focus: Arts and Sciences",
  "Special Focus: Arts, Music, and Design"
), 1, 0),
BUSINESS = ifelse(CARNEGIEAPM %in% c(
  "Special Focus: Business",
  "Special Focus: Applied and Career Studies"
), 1, 0),
STEM = ifelse(CARNEGIEAPM %in% c("Special Focus: Technology, Engineering, and Sciences"), 1, 0)
# leave out CARNEGIEAPM == "Special Focus: Theological Studies"
) %>%
# remove the original categorical variables
select(-CONTROL, -APPFEEUW, -TUITPL1, -TUITPL3, -ADMCON7, -CARNEGIESIZE, -CARNEGIIERSCH, -CARNEGIESAEC)

cat("Number of columns after 1st cleaning =", ncol(private1), "\n")

```

```
## Number of columns after 1st cleaning = 135
```

2nd Cleaning

Check NAs, we don't want to use variables that have a lot of missing rows (over 5%). Variables like Men, Women we will only keep Men. We will also remove other tuition data like out-of-state tuition, published tuition, and other uninterested variables.

##	column	missing	perc_missing
## UNITID	UNITID	0	0.0
## ACTCM25	ACTCM25	422	48.1
## ACTCM50	ACTCM50	422	48.1
## ACTCM75	ACTCM75	422	48.1
## ACTEN25	ACTEN25	452	51.5
## ACTEN50	ACTEN50	452	51.5
## ACTEN75	ACTEN75	452	51.5
## ACTMT25	ACTMT25	451	51.4
## ACTMT50	ACTMT50	451	51.4
## ACTMT75	ACTMT75	451	51.4
## ACTPCT	ACTPCT	336	38.3
## ADMCON1	ADMCON1	142	16.2
## ADMCON2	ADMCON2	142	16.2
## ADMSSN	ADMSSN	142	16.2
## AGRNT_N	AGRNT_N	35	4.0
## AGRNT_T	AGRNT_T	35	4.0
## APPLFEEU	APPLFEEU	497	56.6
## ATHASSOC	ATHASSOC	0	0.0
## AVEDEBT	AVEDEBT	671	76.4
## AVEDEBT_nr	AVEDEBT_nr	668	76.1

## avg_individuals_per_return	avg_individuals_per_return	0	0.0
## CARNEGIEIC	CARNEGIEIC	0	0.0
## CHG2AT3	CHG2AT3	39	4.4
## CHG3AT3	CHG3AT3	39	4.4
## CIPCODE2	CIPCODE2	1	0.1
## CIPCODE3	CIPCODE3	1	0.1
## CIPCODE4	CIPCODE4	1	0.1
## CITY	CITY	0	0.0
## CLASIZ2049	CLASIZ2049	672	76.5
## CLASIZOVE50	CLASIZOVE50	672	76.5
## CLASIZUND20	CLASIZUND20	672	76.5
## COAISGS2	COAISGS2	878	100.0
## COAIST42	COAIST42	878	100.0
## COUNTYCD	COUNTYCD	0	0.0
## COUNTYNM	COUNTYNM	0	0.0
## DISABPCT	DISABPCT	289	32.9
## DVADM01	DVADM01	142	16.2
## DVADM04	DVADM04	144	16.4
## EF2MORT	EF2MORT	1	0.1
## EFAIANT	EFAIANT	1	0.1
## EFASIAT	EFASIAT	1	0.1
## EFBKAAT	EFBKAAT	1	0.1
## EFFT	EFFT	7	0.8
## EFFTMEN	EFFTMEN	1	0.1
## EFFTWOM	EFFTWOM	1	0.1
## EFGNDRKN	EFGNDRKN	196	22.3
## EFGNDRUA	EFGNDRUA	196	22.3
## EFHISPT	EFHISPT	1	0.1
## EFNHPIT	EFNHPIT	1	0.1
## EFNRLT	EFNRLT	1	0.1
## EFTOTAL	EFTOTAL	1	0.1
## EFUG1ST	EFUG1ST	1	0.1
## EFUGCNT	EFUGCNT	1	0.1
## EFUGTRN	EFUGTRN	1	0.1
## EFUNKNT	EFUNKNT	1	0.1
## EFWHITT	EFWHITT	1	0.1
## elderly_share_pop	elderly_share_pop	0	0.0
## ENRLT	ENRLT	143	16.3
## ENRTOT	ENRTOT	1	0.1
## GBA4RTT	GBA4RTT	68	7.7
## GBA6RTT	GBA6RTT	68	7.7
## gdp_per_capita	gdp_per_capita	24	2.7
## GRRTM	GRRTM	64	7.3
## GRRTW	GRRTW	48	5.5
## income_per_capita	income_per_capita	24	2.7
## INSTNM	INSTNM	0	0.0
## LATITUDE	LATITUDE	0	0.0
## LONGITUDE	LONGITUDE	0	0.0
## NPIST2	NPIST2	878	100.0
## PGGRRTT	PGGRRTT	51	5.8
## popestimate	popestimate	0	0.0
## PROPBOR	PROPBOR	672	76.5
## PROPBOR_nr	PROPBOR_nr	668	76.1
## retirement_income	retirement_income	0	0.0

## RMBRDAMT	RMBRDAMT	771	87.8
## RMFRGNCP	RMFRGNCP	23	2.6
## RMINSTTP	RMINSTTP	23	2.6
## RMOUSTTP	RMOUSTTP	23	2.6
## RMUNKNWP	RMUNKNWP	23	2.6
## SATMT25	SATMT25	401	45.7
## SATMT50	SATMT50	401	45.7
## SATMT75	SATMT75	401	45.7
## SATPCT	SATPCT	336	38.3
## SATVR25	SATVR25	401	45.7
## SATVR50	SATVR50	401	45.7
## SATVR75	SATVR75	401	45.7
## SLOB	SLOB	0	0.0
## STATE	STATE	0	0.0
## STUFACR	STUFACR	1	0.1
## TRRTTOT	TRRTTOT	382	43.5
## TUITION2	TUITION2	20	2.3
## TUITION3	TUITION3	20	2.3
## UAGRNT	UAGRNT	1	0.1
## UDGPRNTN	UDGPRNTN	1	0.1
## UFLOANN	UFLOANN	1	0.1
## WEBADDR	WEBADDR	0	0.0
## ZIP	ZIP	0	0.0
## FORPROFIT	FORPROFIT	0	0.0
## ADMCON7_REQUIRED	ADMCON7_REQUIRED	142	16.2
## APPFEEUW_YES	APPFEEUW_YES	1	0.1
## TUITPL1_YES	TUITPL1_YES	1	0.1
## TUITPL3_YES	TUITPL3_YES	1	0.1
## ALLONCAM_YES	ALLONCAM_YES	1	0.1
## ASSOC1_YES	ASSOC1_YES	0	0.0
## BOARD_YES	BOARD_YES	1	0.1
## ROOM_YES	ROOM_YES	1	0.1
## SEMESTER	SEMESTER	0	0.0
## HBCU_YES	HBCU_YES	0	0.0
## HOSPITAL_YES	HOSPITAL_YES	0	0.0
## OPENPUBL_YES	OPENPUBL_YES	0	0.0
## RELAFFIL_NO	RELAFFIL_NO	0	0.0
## SLO6_YES	SLO6_YES	0	0.0
## SLOA_YES	SLOA_YES	0	0.0
## STUSRV2_YES	STUSRV2_YES	0	0.0
## STUSRV3_YES	STUSRV3_YES	0	0.0
## LO_SUBURB	LO_SUBURB	0	0.0
## LO_TOWN	LO_TOWN	0	0.0
## LO_CITY	LO_CITY	0	0.0
## HDEOFR_DOC	HDEOFR_DOC	0	0.0
## HDEOFR_MAS	HDEOFR_MAS	0	0.0
## NORTHEAST	NORTHEAST	0	0.0
## MIDWEST	MIDWEST	0	0.0
## SOUTH	SOUTH	0	0.0
## WEST	WEST	0	0.0
## SZ_MEDIUM	SZ_MEDIUM	0	0.0
## SZ_LARGE	SZ_LARGE	0	0.0
## RSCH_INSTS	RSCH_INSTS	0	0.0
## RSCH_HIGH	RSCH_HIGH	0	0.0

## MED_HI_EARNS	MED_HI_EARNS	0	0.0
## MIXED	MIXED	0	0.0
## PROFESSIONS	PROFESSIONS	0	0.0
## HEALTH	HEALTH	0	0.0
## ARTS	ARTS	0	0.0
## BUSINESS	BUSINESS	0	0.0
## STEM	STEM	0	0.0

Number of columns after 2nd cleaning = 67

3rd Cleaning

We will now do a third time cleaning:

- Remove any non-numerical variables (to remove extra categorical variables we're not interested in. If the number of columns decrease, should check what extra columns got removed). In this case, we removed ATHASSOC (Member of National Athletic Association) because we have one for NCAA already, removed CARNEGIEIC (Institutional Classification) because we have program mix and this variable is over-complicated.
- Remove any row with NAs

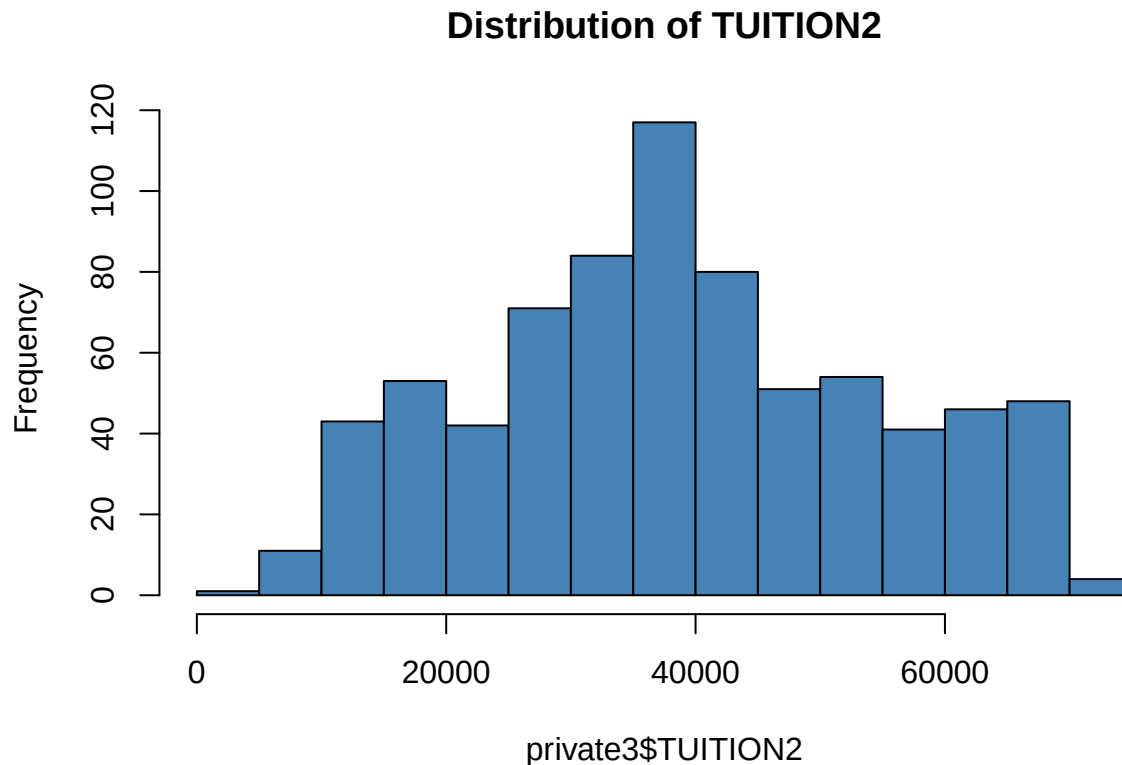
Number of rows after 3rd cleaning = 746

Number of columns after 3rd cleaning = 65

2. Data Transformation

Response Variable: TUITION2

Tuition price is almost always right-skewed and usually need log-transformation but we will check just in case. Here's the distribution of the original TUITION2.



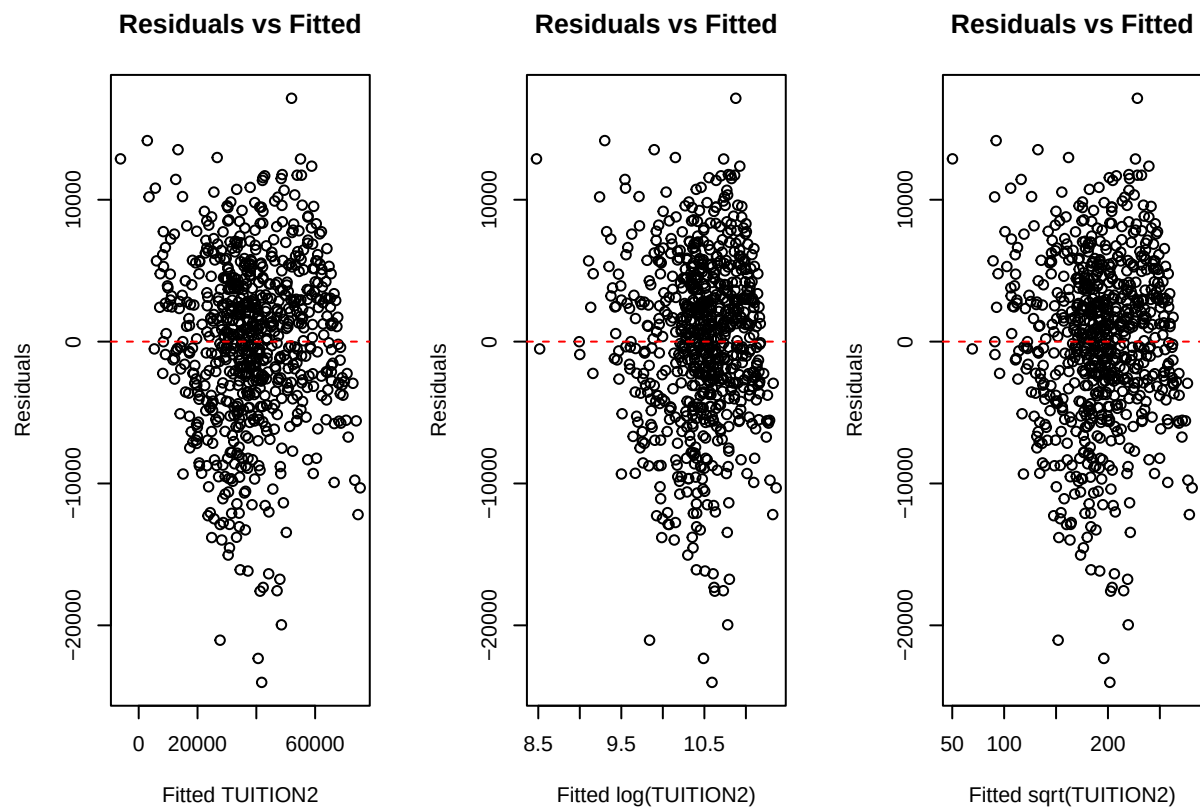
Next, we check the AIC of each model with original TUITION2, log-transformed TUITION2, and sqrt-transformed TUITION2. AIC is the Akaike information criterion, a mathematical method for evaluating how well a model fits the data it was generated from.

```
m1 <- lm(TUITION2 ~ . - UNITID, data = private3)
m2 <- lm(log(TUITION2) ~ . - UNITID, data = private3)
m3 <- lm(sqrt(TUITION2) ~ . - UNITID, data = private3)

AIC(m1, m2, m3)
```

```
##      df      AIC
## m1 65 15204.8151
## m2 65  -186.1078
## m3 65  6407.6650
```

AIC of the model with log-transformed TUITION2 is much lower, so we should use log-transformation. Next, we check Residuals vs Fitted plot for variance behavior.



The residual plot with the **original TUITION2** looks fairly random but:

- Slightly wider spread at higher fitted values
- More extreme negative residuals
- Mild heteroskedasticity (variance increases with fitted values)

The one with **log-transformation**, although has tighter clouds, actually has better variance:

- Spread looks more balanced across fitted values
- Extremes are more symmetric
- Variance looks more stable

The one with **square root transformation** is similar to the original:

- Some widening of spread at higher fitted values
- Slight asymmetry in residuals

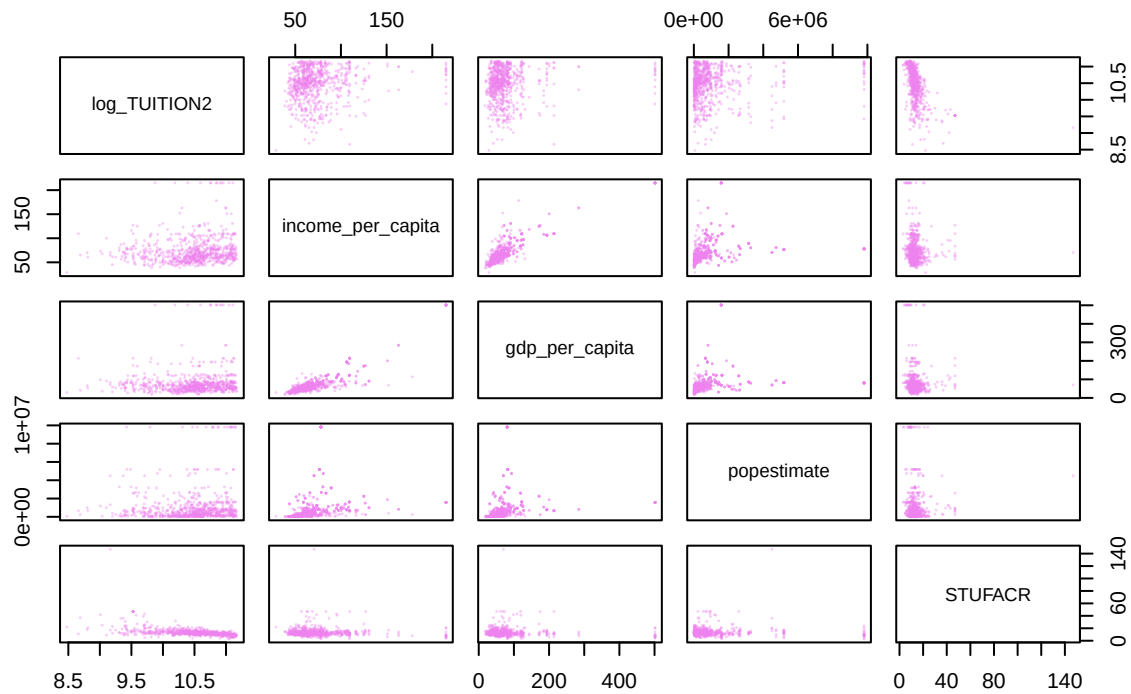
So even though the model with the original TUITION2 gives the highest R^2 , we should choose **log-transformation**. Thus, we add $\log(\text{TUITION2})$ to our dataframe and remove TUITION2 column.

```
private4 <- private3 %>%
  mutate(log_TUITION2 = log(TUITION2)) %>%
  select(-TUITION2)
```

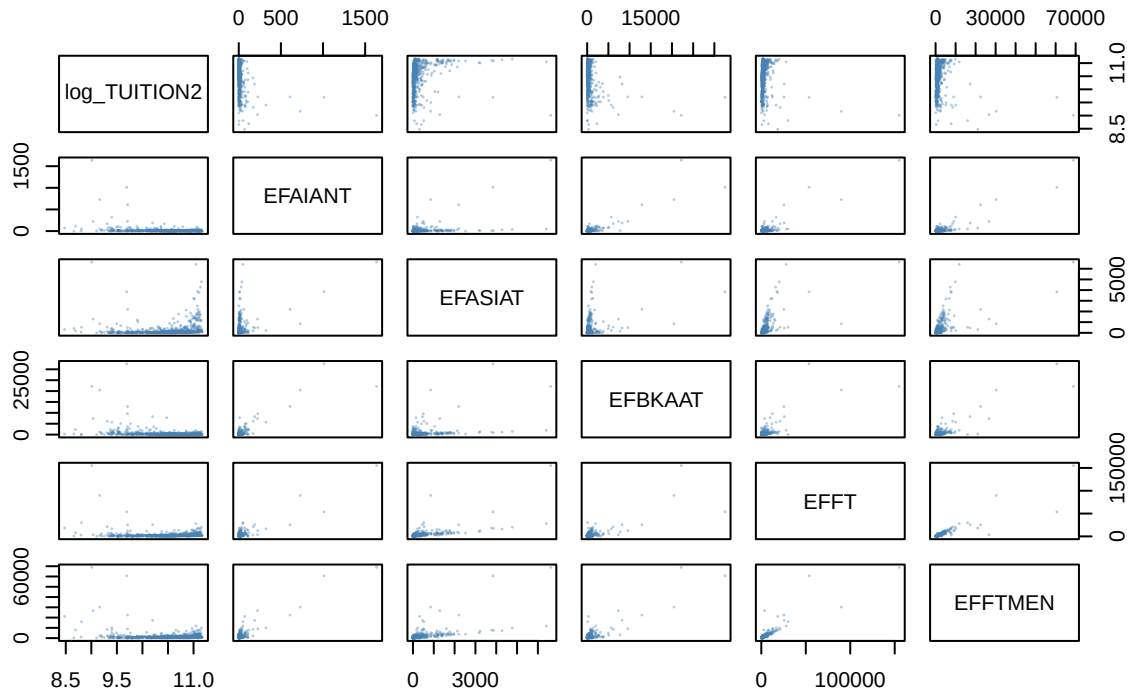
Numeric variables vs log_TUITION2

We will now check numerical variables in pair plots with log_TUITION2 to see if we need to perform transformations.

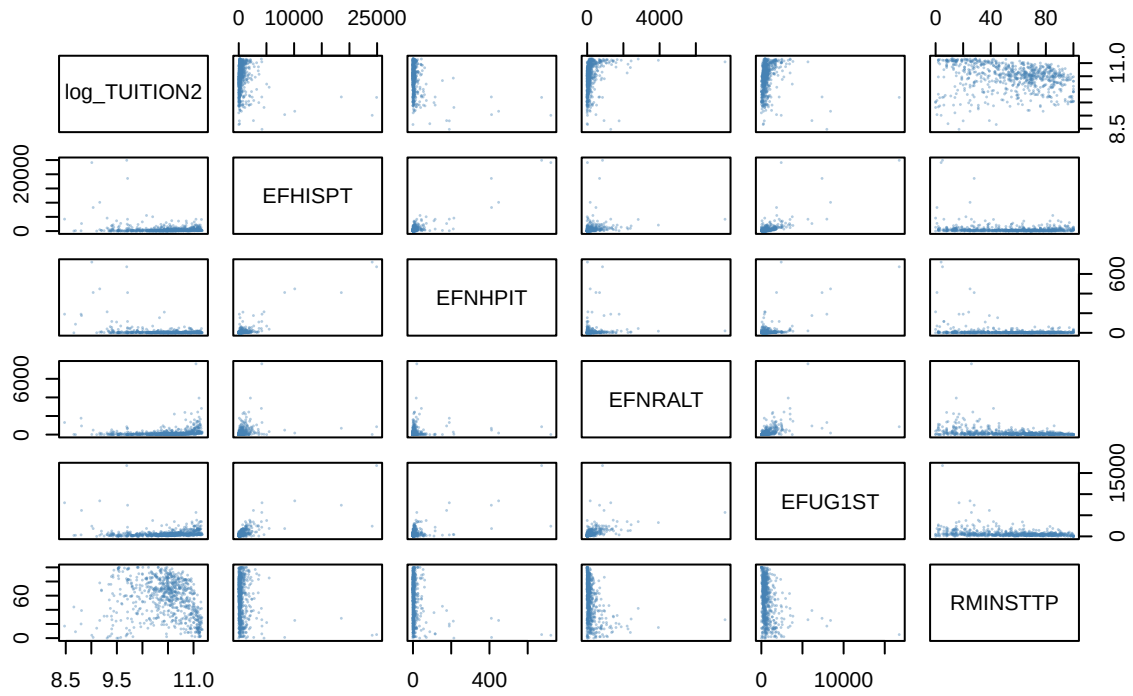
Matrix Plot: Institution Profile & Resources vs In-State Tuition



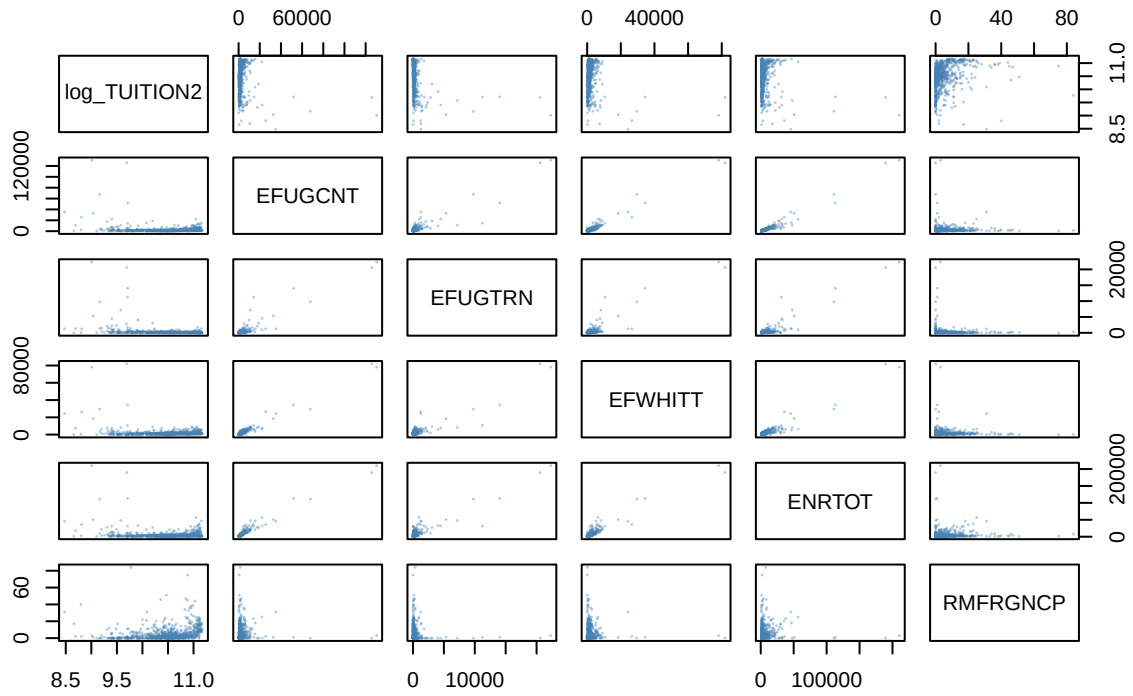
Matrix Plot: Enrollment Variables 1 vs In-State Tuition



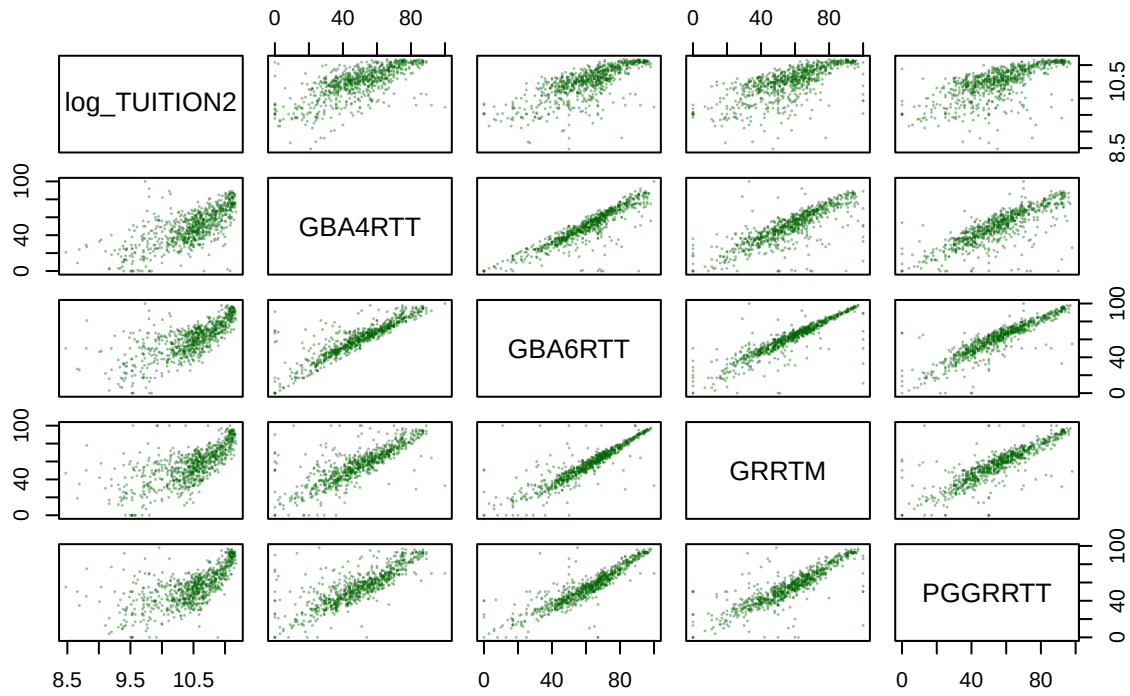
Matrix Plot: Enrollment Variables 2 vs In-State Tuition

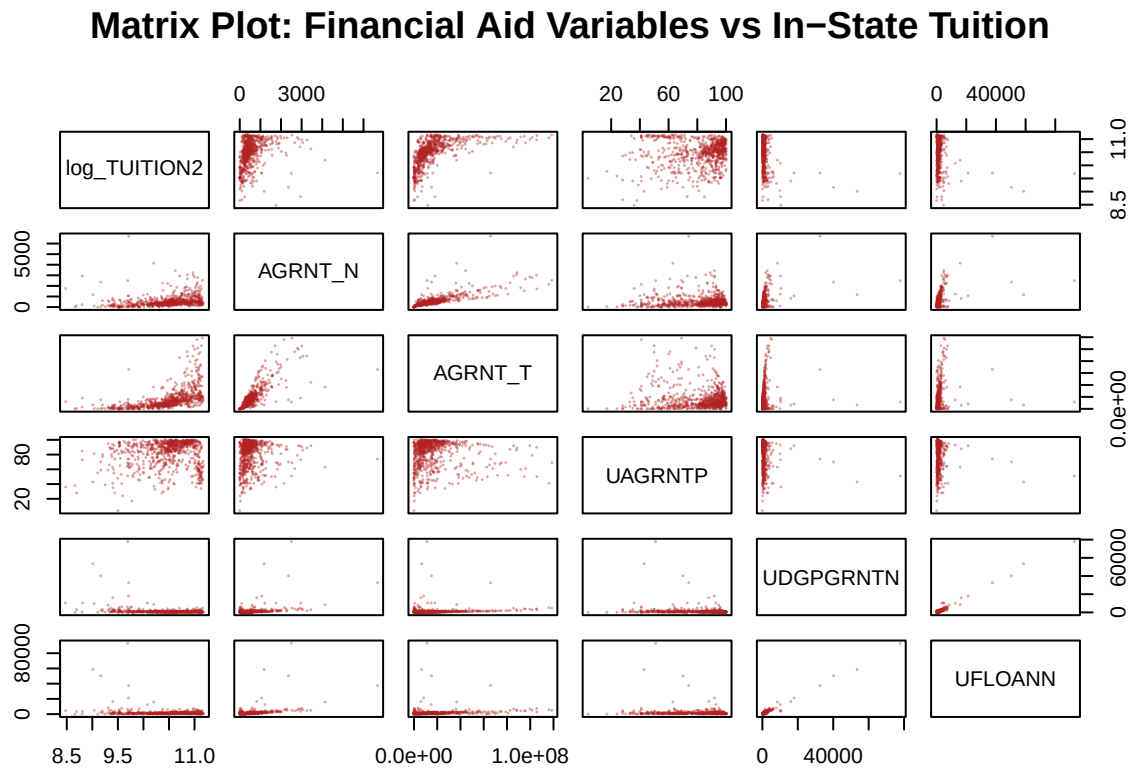


Matrix Plot: Enrollment Variables 3 vs In-State Tuition



Matrix Plot: Admissions & Outcomes vs In-State Tuition





Most of our numerical variables have extreme right skewness. Thus we will:

- Log-transform $\log(x + 1)$ on counts or total dollar amounts (Enrollment & demographic counts, grants, aid and economic resources) since they are heavily right-skewed
- Square-root transformation $\sqrt{x}$ on ratios like STUFACR since it has moderate skewness
- No transformation on variables like graduation rates (GBA4RTT, GBA6RTT, GRRTM, PGRRRTT) and certain percentages (UAGRNT, RMINSTP) since they appear relatively normally distributed or well-spread

```
private5 <- private4 %>% mutate(
  # Log transformations for heavily right-skewed counts and financial totals
  ## Enrollment & Demographic Counts
  log_EFAIANT = log(EFAIANT + 1),
  log_EFASIAT = log(EFASIAT + 1),
  log_EFBKAAT = log(EFBKAAT + 1),
  log_EFFT = log(EFFT + 1),
  log_EFFTMEN = log(EFFTMEN + 1),
  log_EFHISPT = log(EFHISPT + 1),
  log_EFNHPIT = log(EFNHPIT + 1),
  log_EFNRLT = log(EFNRLT + 1),
  log_EFUG1ST = log(EFUG1ST + 1),
  log_EFUGCNT = log(EFUGCNT + 1),
  log_EFUGTRN = log(EFUGTRN + 1),
  log_EFWHITT = log(EFWHITT + 1),
```

```

log_ENRTOT      = log(ENRTOT + 1),
log_RMFRGNCP    = log(RMFRGNCP + 1), # Percentage but heavily right-skewed in plot

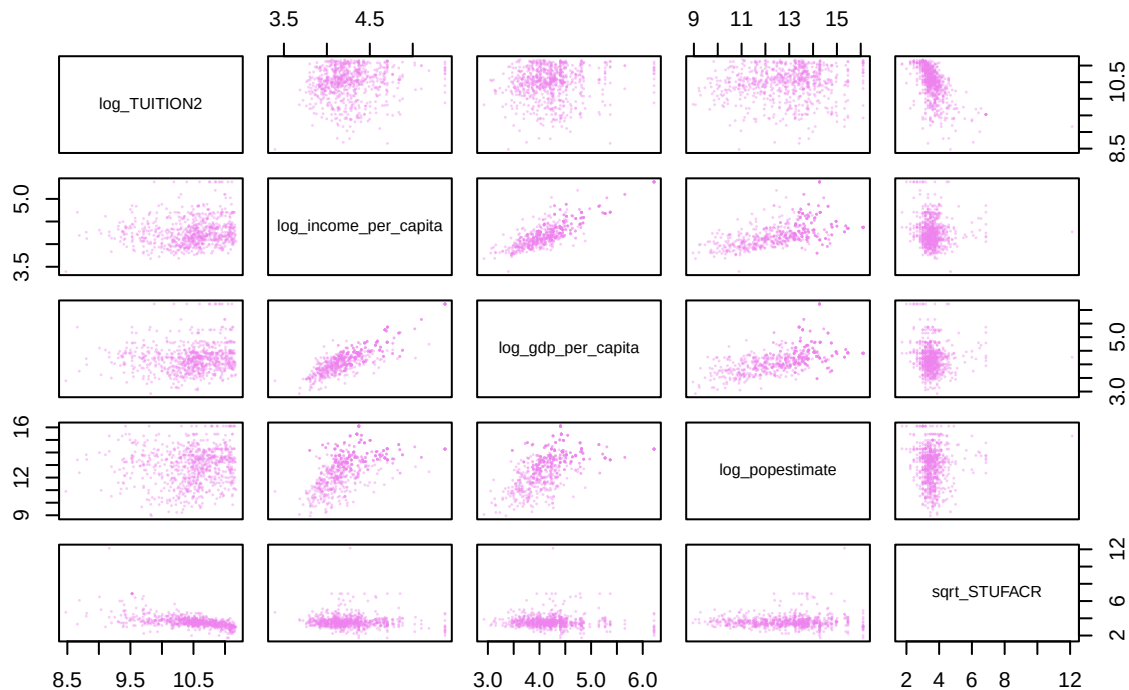
## Financial Aid & Economic Resources
log_AGRNT_N     = log(AGRNT_N + 1),
log_AGRNT_T     = log(AGRNT_T + 1),
log_UDGPGRNTN  = log(UDGPGRNTN + 1),
log_UFLOANN     = log(UFLOANN + 1),
log_income_per_capita = log(income_per_capita + 1),
log_gdp_per_capita  = log(gdp_per_capita + 1),
log_poestimate  = log(poestimate + 1),

# Square Root transformation for moderate skew in ratios
sqrt_STUFACR    = sqrt(STUFACR)
) %>%
# Remove original columns to avoid multicollinearity with transformed versions
select(-EFAIANT, -EFASIAT, -EFBKAAT, -EFFT, -EFFTMEN, -EFHISPT, -EFNHPIT,
       -EFNRALT, -EFUG1ST, -EFUGCNT, -EFUGTRN, -EFWHITT, -ENRTOT, -RMFRGNCP,
       -AGRNT_N, -AGRNT_T, -UDGPGRNTN, -UFLOANN, -income_per_capita,
       -gdp_per_capita, -poestimate, -STUFACR)

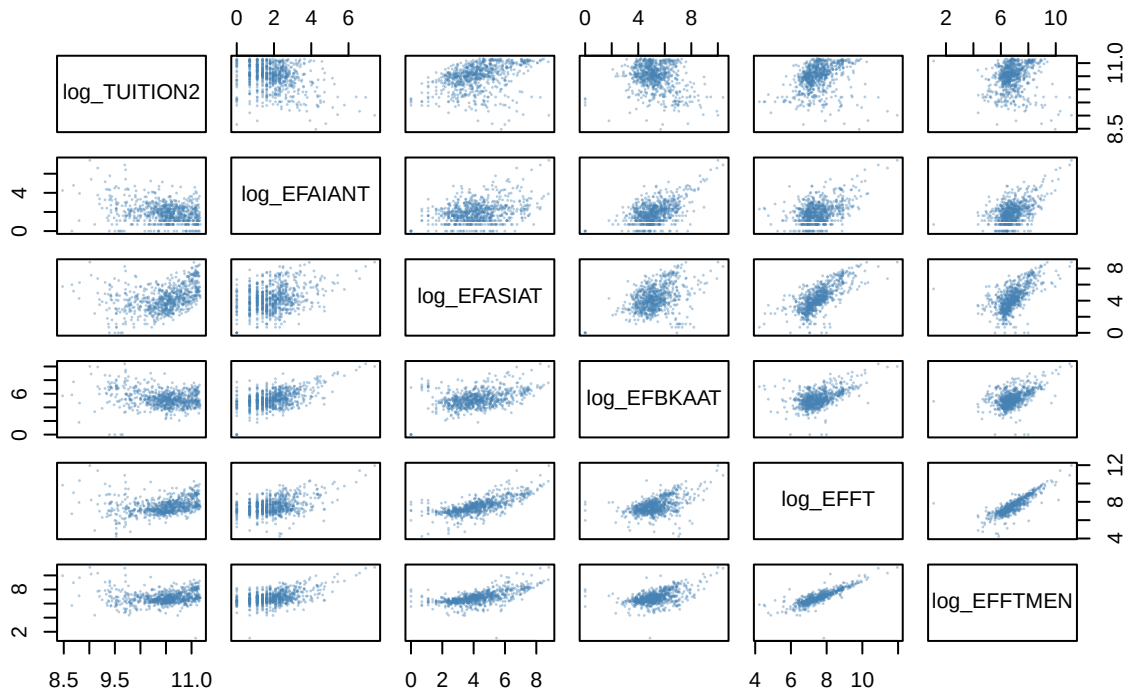
```

The transformed numerical variables pair plots look much better than that of the original numerical variables.

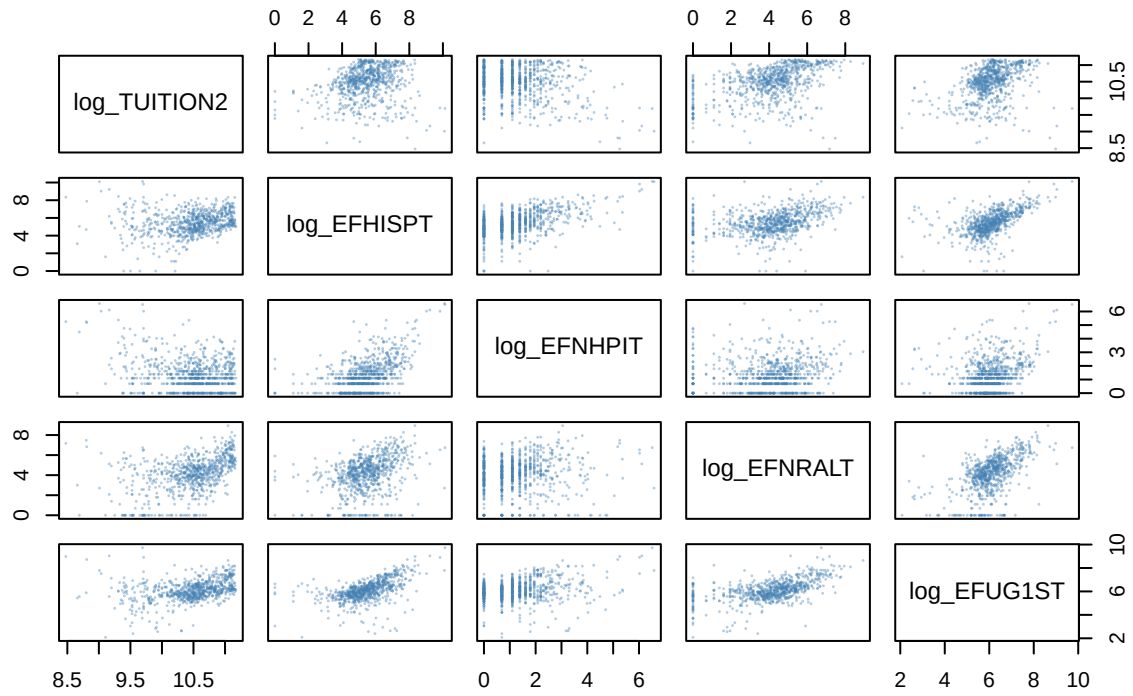
Matrix Plot: Institution Profile & Resources vs In-State Tuition



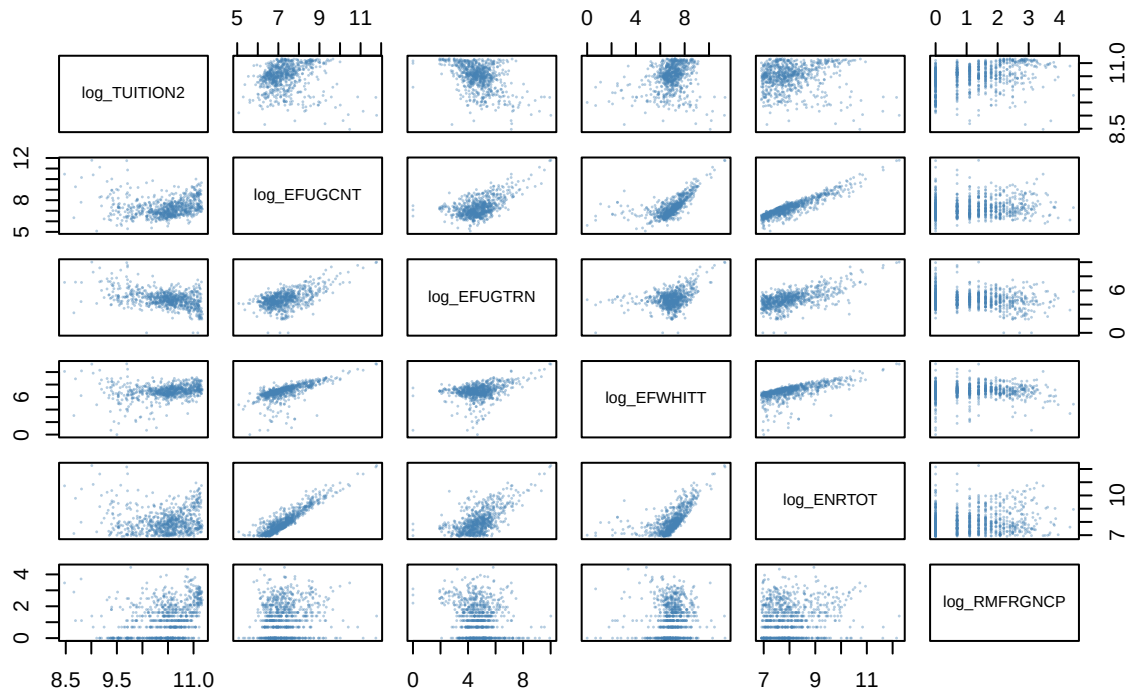
Matrix Plot: Enrollment Variables 1 vs In-State Tuition



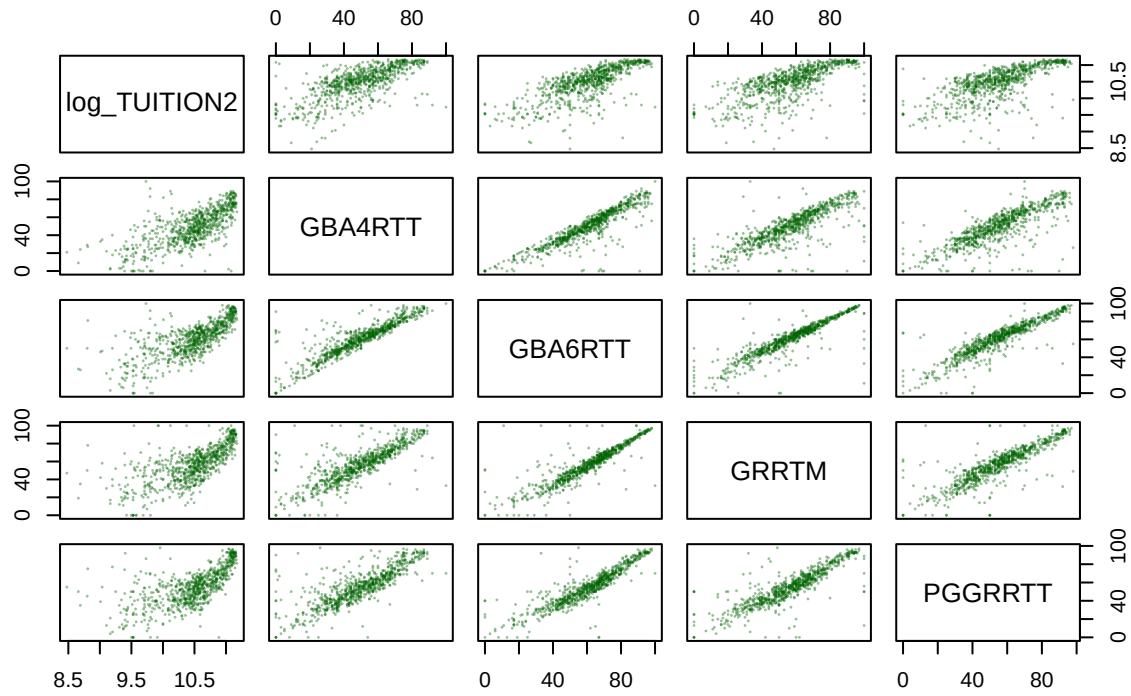
Matrix Plot: Enrollment Variables 2 vs In-State Tuition

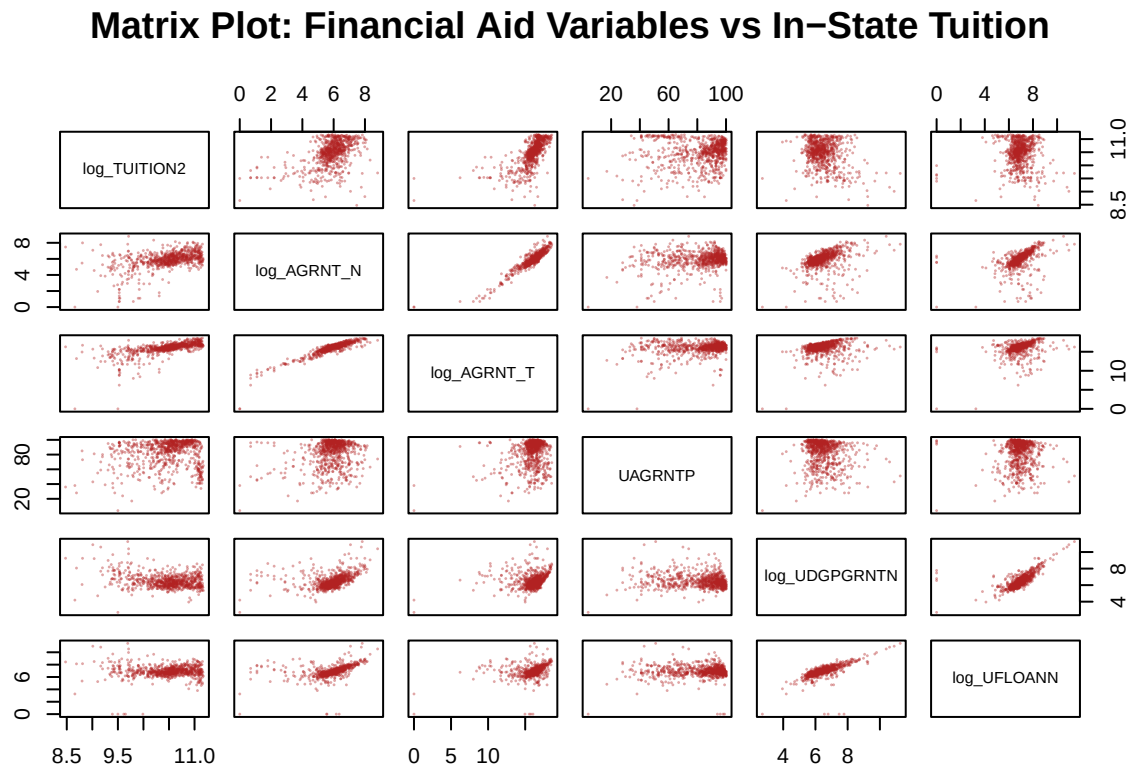


Matrix Plot: Enrollment Variables 3 vs In-State Tuition



Matrix Plot: Admissions & Outcomes vs In-State Tuition





```
## AIC prior to transforming numerical values = -186.1078
```

```
## AIC of after transformations = -234.2514
```

The decrease in AIC also confirmed that the transformations were appropriate.

3. Best Subset Selection

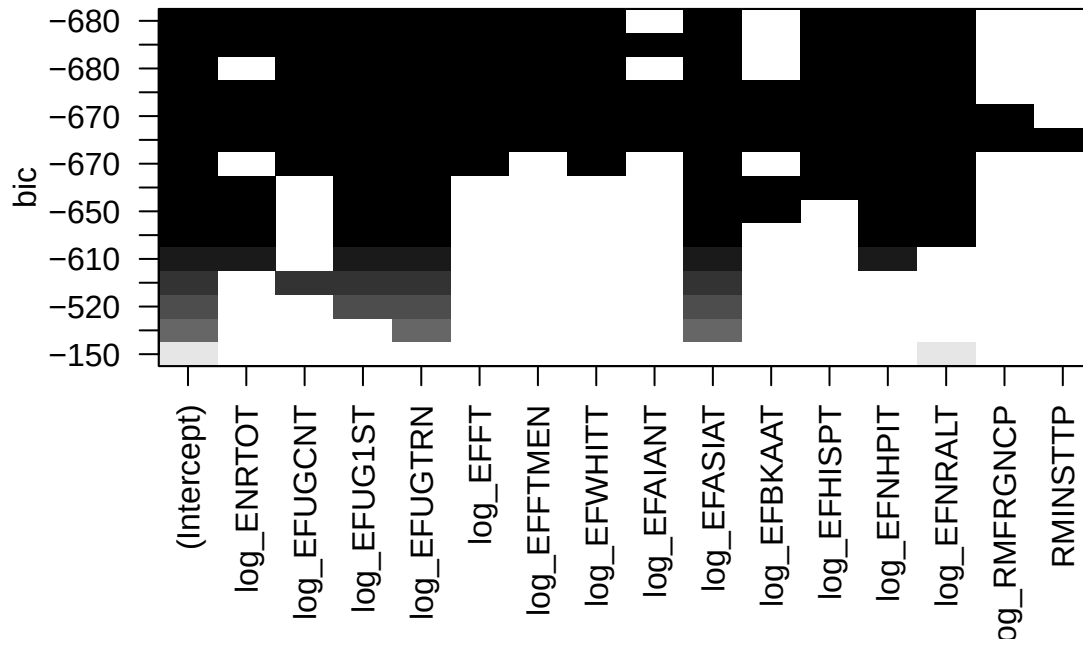
Group Variables into Categories

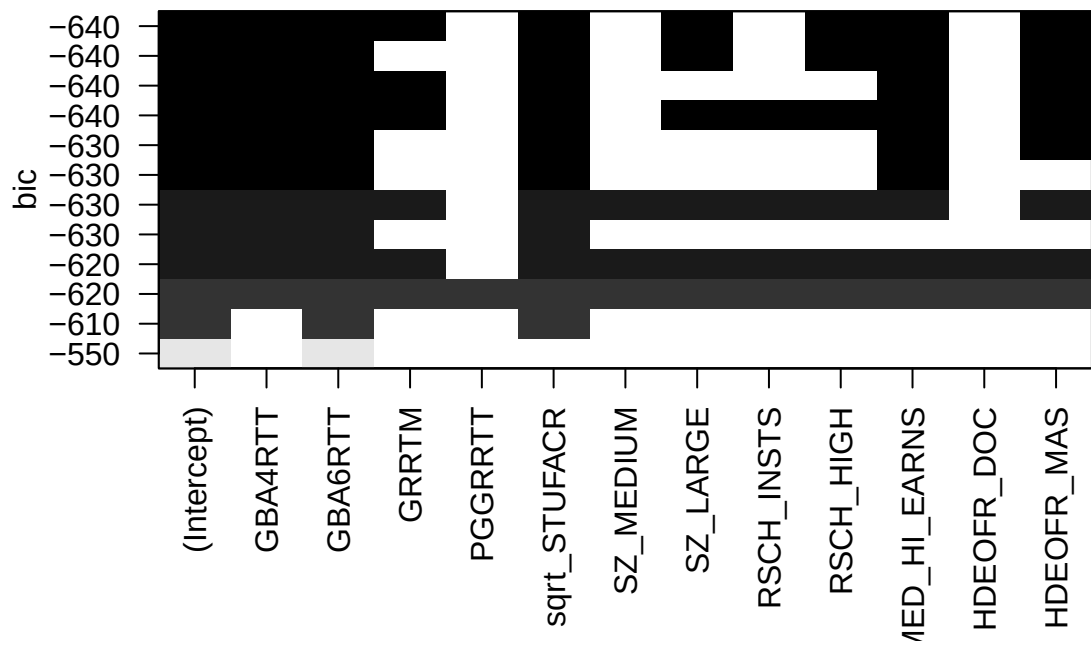
We have too many predictors to run one model. So we split into groups as following (Note that this grouping is strictly for best subset and grouping in the R Shiny App should follow the code book).

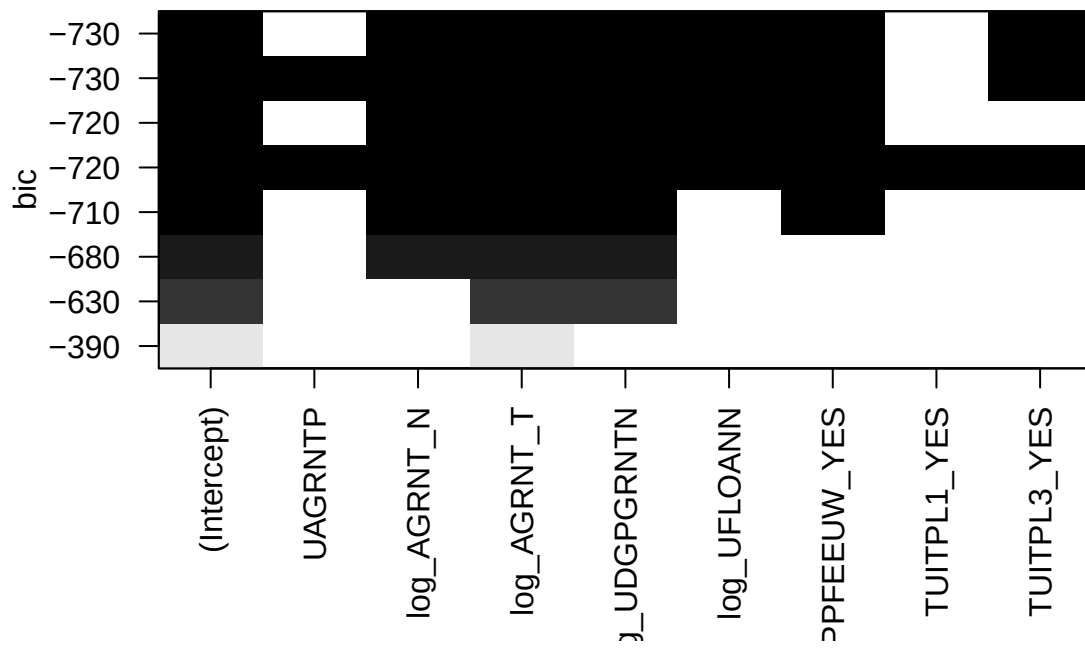
- Group 1: Enrollment & Demographics
- Group 2: Outcomes & Institutional Characteristics
- Group 3: Financial Aid & Grant
- Group 4: Institutional / Structural Indicators
- Group 5: Geography
- Group 6: Program Mix

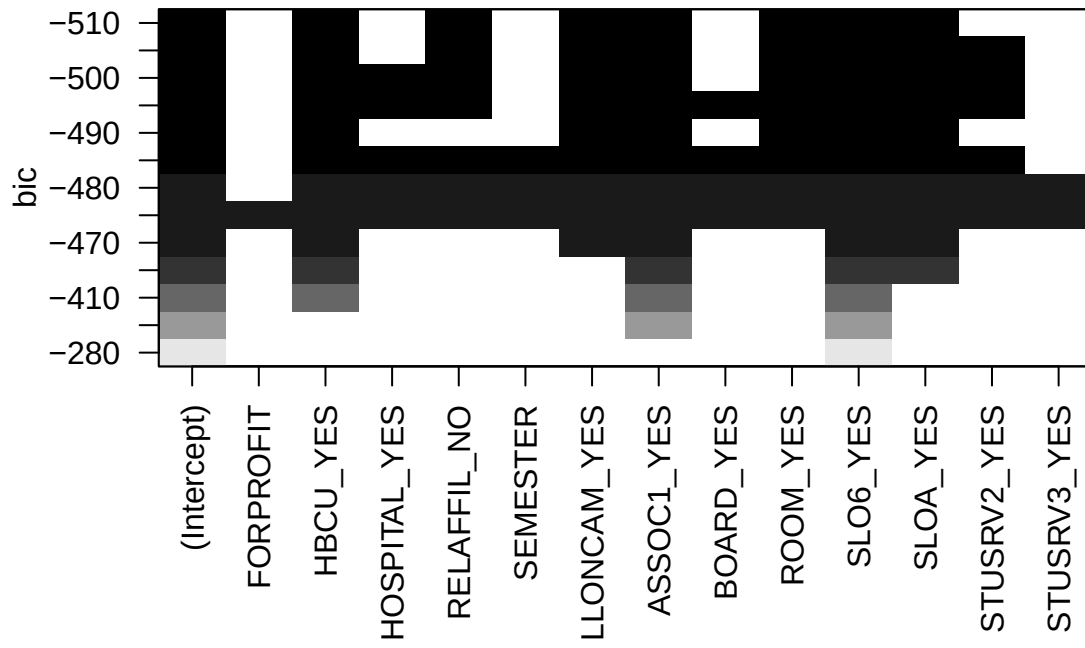

```
## Manual groupings count matches nubmer of columns in private5: TRUE
```

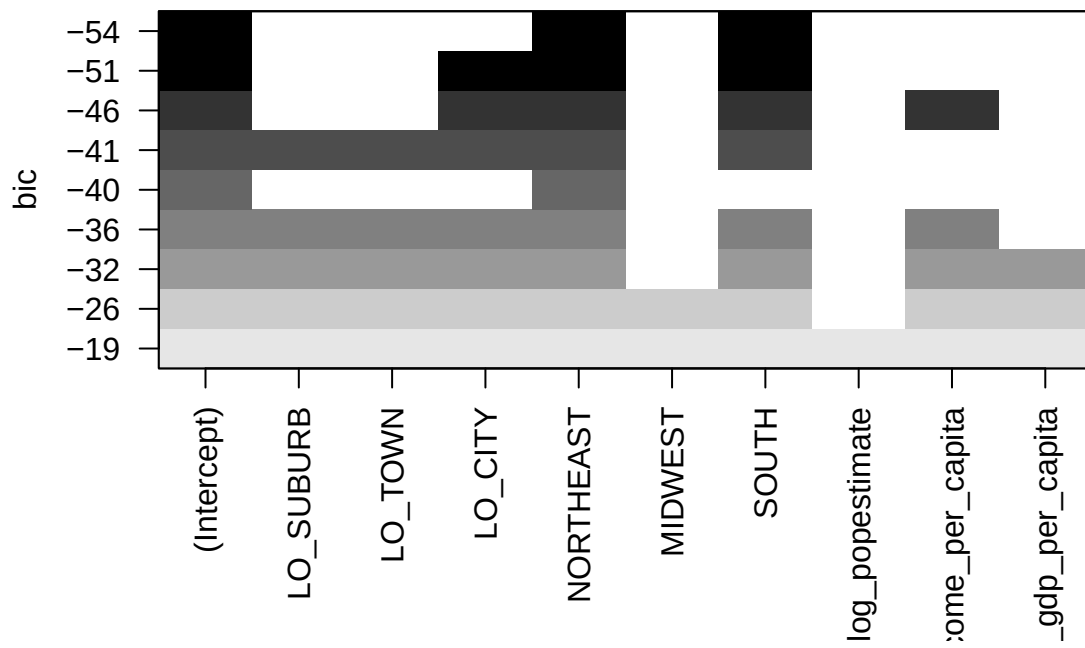
Next, we run best subset on each group and find the best variables from each group:

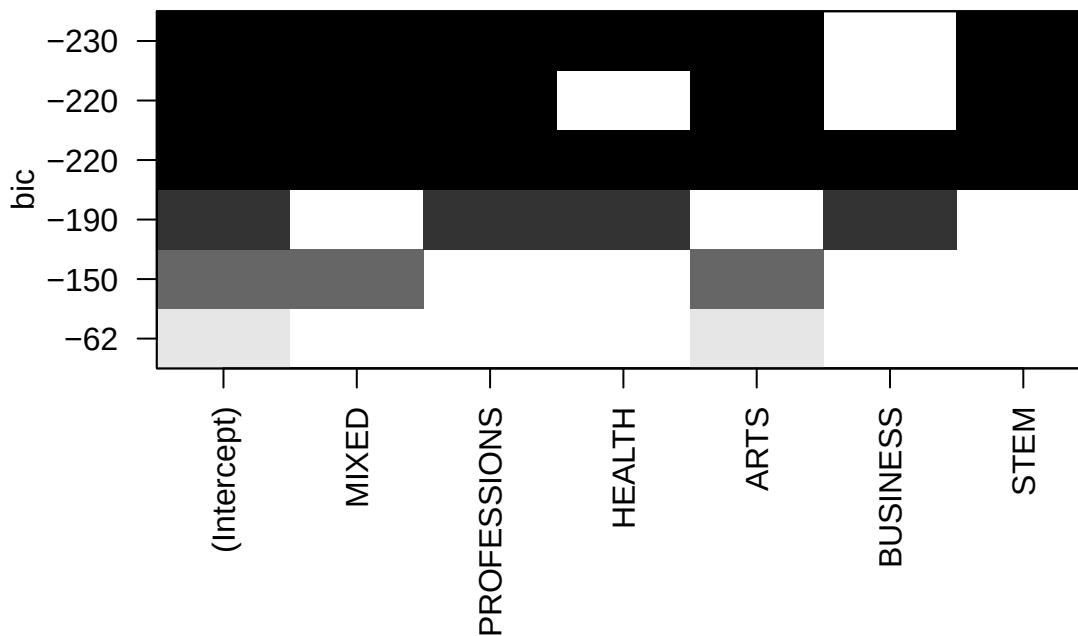












Thus, the most important variables are in the pool (lower bic, better model):

```
## [1] "log_ENRTOT"      "log_EFUGCNT"    "log_EFUG1ST"    "log_EFUGTRN"
## [5] "log_EFFT"        "log_EFFTMEN"    "log_EFWHITT"    "log_EFASIAT"
## [9] "log_EFHISPT"     "log_EFNHPIT"    "log_EFNRLT"     "GBA4RTT"
## [13] "GBA6RTT"         "GRRTM"          "sqrt_STUFACR"   "SZ_LARGE"
## [17] "RSCH_HIGH"       "MED_HI_EARNS"   "HDEOFR_MAS"     "log_AGRNT_N"
## [21] "log_AGRNT_T"     "log_UDGPGRNTN"  "log_UFLOANN"    "APPFEEUW_YES"
## [25] "TUITPL3_YES"     "HBCU_YES"       "RELAFFIL_NO"    "ALLONCAM_YES"
## [29] "ASSOC1_YES"      "ROOM_YES"       "SLO6_YES"       "SLOA_YES"
## [33] "NORTHEAST"       "SOUTH"          "MIXED"          "PROFESSIONS"
## [37] "HEALTH"          "ARTS"           "STEM"
```

Final Best Subset

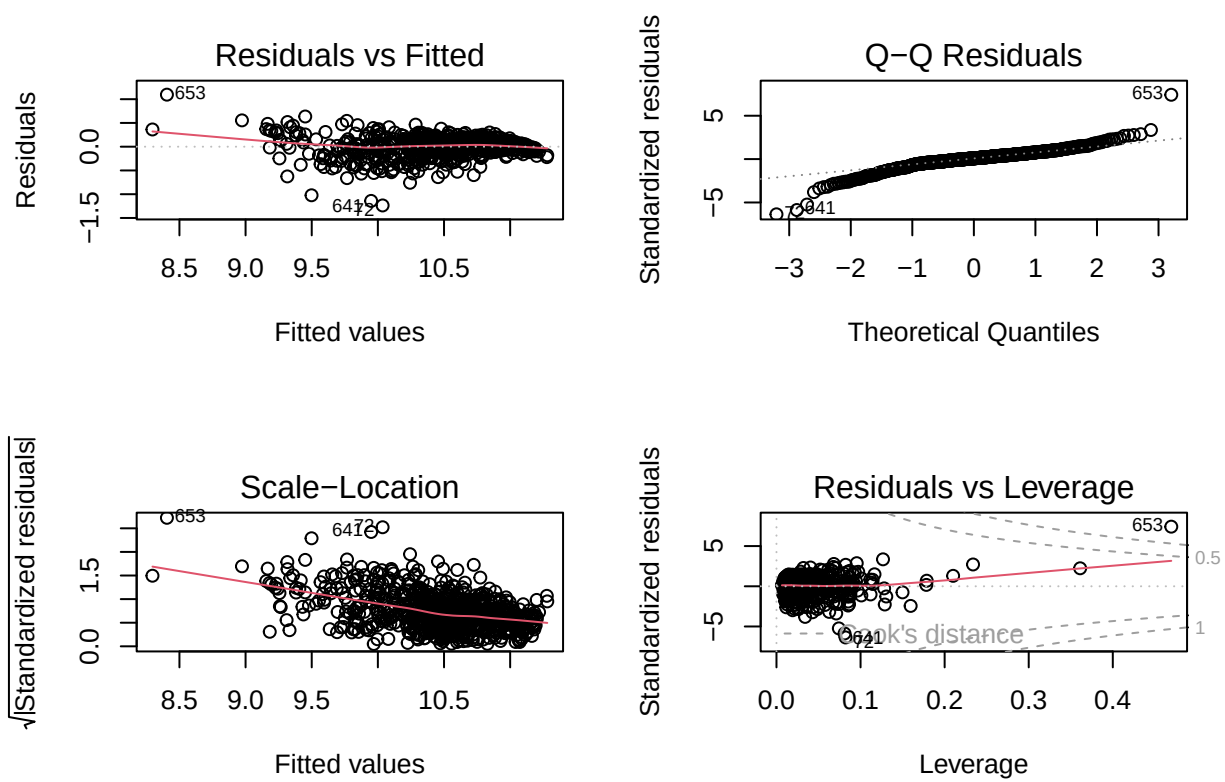
Now, we run another best subset with the variables we deem important from the above analysis.


```

## log_EFHISPT      0.0321271  0.0104514   3.074 0.002192 **
## log_EFWHITT      -0.0915829  0.0146584  -6.248 7.11e-10 ***
## log_EFNHPIT      -0.0423600  0.0091569  -4.626 4.42e-06 ***
## GBA6RTT          0.0013530  0.0009939   1.361 0.173836
## GRRTM             0.0023822  0.0008709   2.735 0.006382 **
## RSCH_HIGH         0.1656396  0.0355769   4.656 3.84e-06 ***
## RSCH_INSTS        0.0388640  0.0317795   1.223 0.221757
## log_AGRNT_N       -0.1997780  0.0226612  -8.816 < 2e-16 ***
## log_AGRNT_T        0.2291621  0.0150137  15.264 < 2e-16 ***
## log_UDGPGRNTN     -0.2351037  0.0258307  -9.102 < 2e-16 ***
## log_UFLOANN        0.0527203  0.0123149   4.281 2.11e-05 ***
## HBCU_YES          -0.6143351  0.0868036  -7.077 3.49e-12 ***
## ASSOC1_YES         0.0790248  0.0205794   3.840 0.000134 ***
## SLO6_YES           0.1823337  0.0319902   5.700 1.75e-08 ***
## MIXED              0.0433172  0.0400597   1.081 0.279918
## HEALTH             0.0257418  0.0436380   0.590 0.555446
## PROFESSIONS        -0.0090893  0.0360933  -0.252 0.801246
## ARTS               0.0809480  0.0423599   1.911 0.056406 .
## STEM              -0.0403897  0.0660010  -0.612 0.540760
## SZ_MEDIUM          -0.0126941  0.0259240  -0.490 0.624521
## SZ_LARGE           -0.0665603  0.0604874  -1.100 0.271525
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2028 on 721 degrees of freedom
## Multiple R-squared:  0.8306, Adjusted R-squared:  0.825
## F-statistic: 147.3 on 24 and 721 DF, p-value: < 2.2e-16

```

We also generate plots to validate the model.



Next step: identifying outliers, validating models, run this for out-of-state tuition, public schools, etc.